

Predicting Neighborhood Gentrification Risk in Washington D.C. Using Supervised Machine Learning

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<https://github.com/FinnRisse/gentrification-risk-prediction>

Abstract

Gentrification and displacement are among the most consequential and least computationally examined problems in contemporary urban planning. In this project, I apply supervised machine learning to census tract-level data for Washington D.C. to predict whether a given neighborhood is at high risk of gentrification. I train and compare Logistic Regression and Random Forest classifiers on a set of housing and socioeconomic features drawn from the BuzzFeed News gentrification dataset. My results show that Random Forest substantially outperforms the linear baseline, achieving an accuracy of 0.851 and an F1-score of 0.681, and that median income and median home value are the strongest predictors of gentrification risk. The findings suggest that even simple machine learning models can give planners early-warning signals before displacement becomes irreversible.

1 Introduction

Gentrification is not a natural disaster; it is a predictable process. The conditions that precede displacement — rising home values, income shifts, demographic change — accumulate over years and leave measurable traces in publicly available data long before longtime residents are priced out. Washington D.C. is one of the most documented cases of this dynamic in the United States; a city whose rapid post-2000 transformation along its transit corridors displaced tens of thousands of longtime Black residents from neighborhoods they had occupied for generations. This project asks whether the census-level signatures of that process can be detected predictively.

I apply two standard classification models to census tract-level data for Washington D.C. to predict which neighborhoods exhibit high gentrification risk, and identify which features drive those predictions.

2 Related Work

Prior work on gentrification prediction using machine learning has found that census-level socioeconomic features are strong predictors of displacement risk. Jayne Yoo explored this approach using American Housing Survey data in the Washington D.C. metropolitan area specifically, finding that a Random Forest classifier achieved 83% accuracy in identifying gentrifying housing units [1]. More recent work has extended these methods to a graph-based deep learning framework applied across Chicago, New York City, and Los Angeles, achieving even stronger precision at the census tract level [2]. A separate body of research has focused on

transit-induced gentrification, finding that neighborhoods near new rail stations experience property value increases significantly faster than comparable areas without transit access [3]; a dynamic that is particularly relevant to Washington D.C., whose Metro system has been directly linked to neighborhood displacement along its corridor.

3 Methodology

3.1 Data and Preprocessing

I used census tract-level data for Washington D.C. from the BuzzFeed News gentrification dataset, which draws on the American Community Survey's 2013–2017 five-year estimates and covers 967 census tracts [4]. BuzzFeed News assembled this data to support a descriptive demographic analysis of gentrification patterns across five U.S. cities; this project builds on that foundation by applying supervised machine learning to the same raw census features — an analytical step their work did not take. The dataset does not include a pre-built gentrification label, so I derived one following the demographic-shift approach standard in the gentrification literature: a tract was labeled high-risk if it fell above the median in both educational attainment and white population share, and below the median in Black population share simultaneously — a combination that captures the classic demographic signature of gentrification in D.C. specifically. Housing and income features were then used as predictors, kept strictly separate from the label construction to prevent data leakage. The dataset was split 80/20 into training and test sets using stratified sampling.

3.2 Models

I trained two models. **Logistic Regression** is the simplest baseline, predicting risk as a weighted combination of input features. **Random Forest** builds many decision trees on random subsets of the data and averages their predictions, allowing it to capture more complex, nonlinear relationships between features that a linear model would miss.

4 Results

Table 1 shows test set performance for both models. Random Forest substantially outperforms Logistic Regression on both metrics. The weak F1-score of Logistic Regression reflects its difficulty handling the class imbalance between high- and low-risk tracts under a linear decision boundary; Random Forest handles this considerably better.

Model	Accuracy	F1-Score
Logistic Regression	0.716	0.203
Random Forest (best)	0.851	0.681

Table 1: Test set performance on Washington D.C. census tracts. Best results in bold.

The most important features identified by Random Forest were median income (0.308), median home value (0.284), share of population over 25 (0.156), total population (0.130), and Hispanic

or Latino share (0.122). This is consistent with what the planning literature would predict; neighborhoods where income and home values are shifting fastest are structurally the most exposed to speculative investment pressure and the displacement that follows.

5 Conclusion

This project demonstrates that a Random Forest classifier can identify neighborhoods at high gentrification risk in Washington D.C. with meaningful accuracy using publicly available census data, substantially outperforming a linear baseline. Median income and home value are the most consistent predictors. The broader point is not a technical one; data science tools, when pointed at the right problems, can support more proactive and equitable planning decisions. Washington D.C. has already lived through the consequences of transit investment without displacement protection. Other cities still navigating that same inflection point could benefit from tools like this one — but only if the political will to act on them exists.

References

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- [3] Dong, H. (2017). Rail-transit-induced gentrification and the affordability paradox of TOD. *Journal of Transport Geography*, 63, 1–10.
- [4] BuzzFeed News. (2020). *Gentrification and Demographic Analysis Dataset*. GitHub.